DULAKSHI GAMMANPILA

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Final-year Computer Science undergraduate with hands-on experience in full-stack software development, backend engineering, and AI-powered applications. Proficient in Python, JavaScript, TypeScript, Java, PHP, Spring Boot, Node.js, FastAPI, React, Next.js, REST API development, and SQL databases, with experience in cloud technologies, Docker, Git, and Agile development. Experienced in developing Generative AI applications using Large Language Models (LLMs), LangChain, Hugging Face, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Embeddings, Semantic Search, and Vector Databases. Passionate about solving real-world problems through research, rapid prototyping, scalable software solutions, and continuous learning, with strong problem-solving, collaboration, and software engineering skills.

EDUCATION

Bachelor of Computing (Honours) in Computer Science
University of Sri Jayewardenepura (Expected Graduation: 2027)

TECHNICAL SKILLS

Programming Languages:
Python, Java, JavaScript, TypeScript, PHP, SQL

Core Concepts:
Object-Oriented Programming (OOP), REST API Development, Agile Development

Web Development:
Frontend: React, Next.js, HTML5, CSS3, Tailwind CSS, Bootstrap
Backend: Spring Boot, Node.js, FastAPI, RESTful APIs

Mobile Development
React Native

Machine Learning:
PyTorch, Hugging Face Transformers, Scikit-learn, XGBoost, SHAP, Pandas, NumPy, OpenCV

Generative AI:
Gemini API, Groq API, LangChain, Prompt Engineering, Retrieval-Augmented Generation (RAG), Vector Embeddings, Large Language Models (LLMs)

Databases & Vector Stores:
MySQL, PostgreSQL, Supabase, ChromaDB

Cloud, DevOps & Tools:
Docker, Azure Container Apps, Vercel, Git, GitHub Actions, CI/CD

PROJECTS

NeuraScan - Brain Tumor MRI Classifier
Technologies: Python, FastAPI, React, Tailwind CSS, EfficientNetB0, ViT-Base, PyTorch, Hugging Face Transformers, Azure Container Apps, Docker, GitHub Actions
• Fine-tuned EfficientNetB0 and Vision Transformer models on MRI scan data, achieving around 94% test accuracy.
• Implemented data preprocessing, augmentation, model evaluation, and performance comparison using PyTorch.

• Built a FastAPI backend serving the model from Hugging Face, containerized both services with Docker, and automated deployment to Azure Container Apps via GitHub Actions CI/CD.
Live demo: https://neurascan-frontend.mangoforest-74886e44.southeastasia.azurecontainerapps.io/
GitHub: github.com/Dulakshi-dev/neurascan

MediScan - AI Medical Report Analyzer
Technologies: Python, LangChain, FastAPI, Next.js, TypeScript, RAG, Supabase, ChromaDB, Groq API, Hugging Face Transformers, Vercel
• Built a custom PDF parser using PyMuPDF to extract lab values from reports with no standard layout.
• Implemented a RAG pipeline (LangChain, ChromaDB, Hugging Face sentence-transformers) with Groq's Llama 3.1 8B to generate grounded, plain-English explanations from retrieved context.
• Deployed backend (FastAPI + Supabase) on Hugging Face Spaces via Docker; frontend (Next.js + TypeScript) on Vercel.
Live demo: https://mediscan-iota-two.vercel.app/
GitHub: github.com/Dulakshi-dev/MediScan

ParkEase - Microservices Parking Reservation System
Technologies: Java, Spring Boot, Spring Security, Spring Cloud Gateway, PostgreSQL, JWT, Docker, Docker Compose, JUnit, Mockito, OpenAPI/Swagger
• Designed a microservices backend with independent Driver and Reservation services, each with its own PostgreSQL database, communicating via authenticated REST calls and unified through a Spring Cloud Gateway.
• Implemented JWT authentication with BCrypt password hashing for users, plus a separate API-key authentication layer for internal service-to-service communication.
• Containerized all services and databases with Docker Compose, covered core business logic with JUnit/Mockito unit tests and documented APIs using OpenAPI/Swagger and a tested Postman collection.
GitHub: https://github.com/Dulakshi-dev/parkease

Shelf Loom - Library Management System
Technologies: PHP, MySQL, JavaScript, Bootstrap, AJAX, MVC, RBAC
• Developed a full-stack library management system with role-based access control (RBAC) and AJAX-based interactions.
• Led a six-member Agile team as Project Manager, coordinating sprints and contributing to database design and testing.
GitHub: github.com/Dulakshi-dev/LMS

VerdictAI - AI Powered Multilingual Legal Assistance App (In Progress)
Technologies: Next.js, TypeScript, Tailwind CSS, RAG, Supabase, Vector Embeddings, Gemini 2.5 Flash, Prompt Engineering, next-intl
• Building an LLM-powered multilingual legal assistant for Sri Lankan law (Sinhala, Tamil, English) using RAG with Supabase vector embeddings and Gemini 2.5 Flash.
• Designed prompt-engineering workflows to improve answer relevance and reduce hallucinations.
GitHub: github.com/ai-codeforge/legal-assist

VOLUNTEERING

• Design Standing Committee Member, IEEE WIE Affinity Group, University of Sri Jayewardenepura (2025)

REFERENCES

• Dr. S.P.B.M.Senadheera, Senior Lecturer, Department of Information Systems Engineering and Informatics, Faculty of Computing, University of Sri Jayewardenepura
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